

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 30%

QUESTIONS: 05

Total Marks:50

Annexure A MOCK INTERVIEW

<i>Criteria</i>	Technical Knowledge (2.5)	Problem Solving (2.5)	Analytical Thinking (1.5)	<i>Communication(1.5)</i>	Confidence (1)	Response to Questions(1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	15%	15%	10%	10%	100%
<i>Q1</i>							
<i>Q2</i>							

Code of Conduct:

I S. Blessika(192525251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student. S.Blessika

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

(Each question aligned with BL3–BL4, 0.75 Marks each)

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example. *(BL3 – 0.75 Marks)*

Q2. Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used. *(BL4 – 0.75 Marks)*

Q3. Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B

TID Items

T4 A, B

T5 A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining. (BL3 – 0.75 Marks)

Q4 . Dataset: Multilevel Data

TID Items

T1 Electronics, Mobile

T2 Electronics, Laptop

T3 Electronics, Mobile

T4 Electronics, Mobile

T5 Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional rules. (BL4 – 0.75 Marks)

Q5. Dataset: Large Transaction Sample

TID Items

T1 A, B, C

T2 A, B

T3 B, C

T4 A, C

T5 A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues. (BL4 – 0.75 Marks)

